

CMPE442 Programming Assignment 3 Report

Tuna Bekiş

11404148014

Introduction

This report shows the results of implementing K-Means and Gaussian Mixture Models from scratch and applying image segmentation to 4 different images from the COCO2017 validation set. Algorithms were tested for K values 2, 3, 5 and 8 for the target number of clusters and the acquired segmentation maps were compared according to computational performance and image quality.

Algorithmic Approach and Theory

-K-Means (Hard Clustering):

K-means perform hard assignment by assigning each pixel to one cluster according to the Euclidean distance to the centroids. The algorithm essentially assumes clusters are spherical in space and have the same variance value.

-GMM (Soft Clustering):

On the other hand, GMM Works with the Expectation-Maximization algorithm and calculates the probability of each pixel belonging to each cluster. Thanks to the covariance parameter, GMM can model not only spherical distributions, but also elliptical distributions.

Comparative Visual Analysis

- Impact of K Values:

On low K values such as 2 or 3, both algorithms converted different objects into color blocks (under-segmentation) and caused detail loss. For K values 5 and 8, textures and object borders became significantly sharper.

-K-Means vs GMM:

The strict distance calculations of K-Means algorithm created sharp, blocked and artificial borders especially in regions with light changes and color gradients. GMM, on the other hand, was able to segment shadows, reflections and color gradients much more naturally and smoothly, thanks to its probabilistic and flexible structure.

Computational Performance

Two algorithms showed opposition in terms of speed. K-means, since it does basic distance computation, converged fast. However, GMM showed a computationally expensive structure due to having to do multivariable normal distribution and matrix inversion computations on tens of thousands of pixels for both E-step and M-step, providing a significantly longer runtime than K-Means.

Visual Appendix (Segmentation Maps)

Photo1:

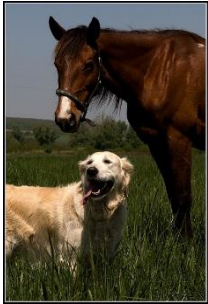


Photo1 Segmentation Map:



Photo2:



Photo2 Segmentation Map:

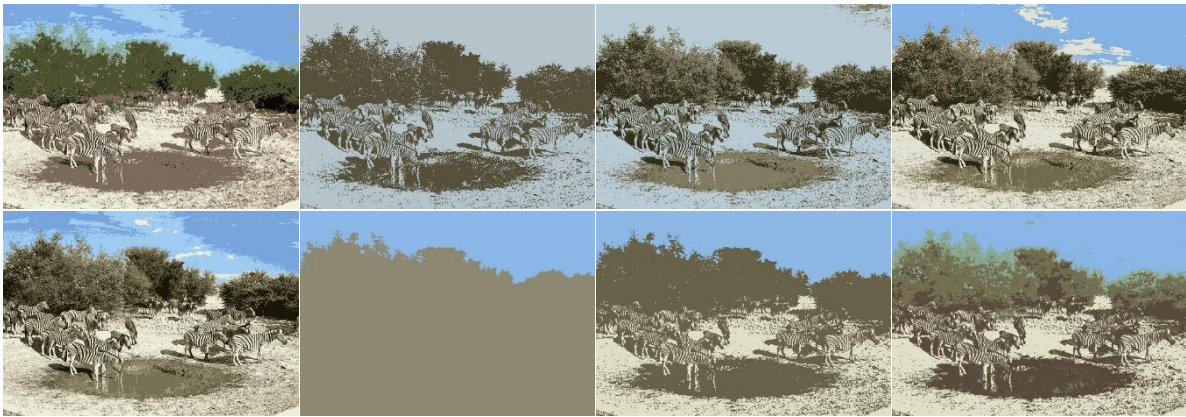


Photo3:



Photo3 Segmentation Map:



Photo4:



Photo4 Segmentation Map:

